Geological and Geophysical Implications of Paleomagnetic Results from Africa

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Abstract. A critical review of all African paleomagnetic results has yielded 31 sets of data that are useful for discussions of the behavior of the paleomagnetic field and of polar wandering and continental drift. Most of the reliable data are new. They largely confirm the previously published polar-wander path relative to Africa for Permian and younger time and add detail particularly about rates of polar shift relative to that continent. They permit a tentative extension of the polar-wander path back to about 600 m.y. (beginning of Cambrian time). A complete revision of the previously published Precambrian polar-wander curve is presented in two parts, covering the periods from 2700 m.y. to about 2100 m.y. and from 2000 m.y. to 1300 m.y., respectively. For these two periods it is possible to use paleomagnetism as a tool in geological correlation and as a means of dating. There is still no information on the interval from about 2100 m.y. to 1950 m.y. or for the interval between 1300 m.y. and 600 m.y. The Mesozoic data remain consistent with Du Toit's reconstruction of Gondwanaland for the Jurassic. The Permo-Carboniferous data confirm the period of rapid drift within the Carboniferous, which has already been identified in Australia by Irving, and confirm the general correctness of the relative positions of Australia and Africa suggested by L. C. King. Comparison of African data with data for the Canadian shield for the period 2000 to 1200 m.y. enables us to suggest tentatively that the Canadian and Southern African cratons did not move relative to each other during that time. This is the first attempt at reconstruction of the relative positions of two continental regions from Precambrian paleomagnetic data. The oldest Precambrian data confirm the existence of a terrestrial magnetic field which occasionally reversed itself at least 2700 m.y. ago. The dispersion of directions in reliable paleomagnetic data is taken as a measure of paleosecular variation; only 3 of the 19 estimates are significantly different from the secular variation associated with the present geomagnetic field.

INTRODUCTION

In the three years since the last review of paleomagnetic data from Africa [*Gough et al.*, 1964*b*] a considerable amount of new information has been ob-

tained, mainly from work carried out at the University College of Rhodesia under grant GP 2533 from the National Science Foundation. The comprehensive lists of data given here comprise 17 entries from Precambrian rocks (Table 1) and 69 entries from Phanerozoic rocks (Table 2). Of these 86 entries only 26 were listed by *Irving* [1964] and only 19 were reviewed by *Gough et al.* [1964*b*]. They are believed to be a thoroughly up-to-date summary of the present state of paleomagnetic knowledge of Africa. The results make up a highly inhomogeneous body of data. Some entries are based on detailed sampling, some on reconnaissance surveys. Only a few entries cover studies which include field and laboratory tests of the stability of remanent magnetization. It is important that in interpreting paleomagnetic results the most reliable data should be recognized, and therefore two large appendixes to this paper are devoted to a critical summary of each result and an assessment of its reliability.

Thirty-one sets of data emerge from this review as being thoroughly reliable and useful in discussions of the behavior of the paleomagnetic field and of relative movements of the continents and poles. The main body of the paper consists of brief discussions of these problems in the light of the African data.

SUMMARY OF AFRICAN PALEOMAGNETIC RESULTS

Tables 1 and 2 and Appendixes 1 and 2 are modeled on the compilations of *Irving* [1964] from which much of the explanation and notation has been taken. Results which are not considered fully reliable are entered in brackets. Some results which do not, by themselves, satisfy the criteria for reliability are allowed to stand either because they can be combined with results of other studies on the same rock group to make up a reliable result or because they serve to confirm results on other formations of similar age.

Column 1 contains reference numbers to the mean pole position. These reference numbers are assigned only to the poles from reliable studies (or a combination of reliable studies); Precambrian poles are prefixed with the letter P and Phanerozoic poles with the letter B to conform with the notation of *Briden* [1967*b*], who lists similarly reliable poles from other continents.

Column 2 contains reference numbers to the appendix.

Column 3 specifies the rock unit, its locality, latitude, and longitude.

Column 4 gives the age of the rock unit studied. The geological periods are abbreviated thus: Precambrian P€, Cambrian €, Ordovician O, Silurian S, Devonian D, Carboniferous C, Permian P, Triassic Tr, Jurassic J, Cretaceous K, Tertiary and Quaternary T. The Cambrian to Cretaceous are subdivided into upper u, middle m, and lower l, where recognized. The Tertiary and Quaternary are subdivided using suffixes e (Eocene), o (Oligocene), m (Miocene), p (Pliocene), and pl (Pleistocene); the Recent is denoted by R. If radioactive age determinations are available, the estimates are indicated by an asterisk, and references and explanations are given in the notes. Otherwise the absolute age is estimated by reference to geological evidence and *Kulp*'s [1961] time scale.

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TABLE 1. Paleomagnetic Results for the Precambrian of Africa

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										Paleomagnetic Pole						
Pole No.	Appendix Reference No.	Rock Unit and Location	Age, m.y.	D	I	k	α_{95}	Δ	Polarity	Latitude	Longitude	K	A_{95}	dp, dm	Pole Quoted	Reference *Irving* [1964]
P1	1 1	Modipe gabbro, Botswana, 24.5S 26 0E	P€ [2670 ± 470]*	154½	+85	80	5½	$9.1^{+4.4}_{-3.0}$	M	33S	31E	22	10½		N	
P2	1 2	Great Dyke, Rhodesia, 18 5S 30.5E	P€ [2530 ± 30]*	218½	−58½	68	6	$9\ 8^{+5.2}_{-3.4}$	−	21½N	61½E	33	9		S	1.63
	1 3	Great Dyke, Rhodesia, 18.0S 30.5E	P€ [2530 ± 30]*	236½	−61	12	20		−	10½N	69E			24, 31	S	1.64
P3	1 4	Gaberones granite, Botswana, 25.0S 25.5E	P€ [2340 ± 50]*	55½	−6½	11	18	$24\ 4^{+15.7}_{-9.5}$	M	34½N	103½E	15	16		N	
	1 5	Gaberones granite, Botswana, 25 0S 25 5E	P€ [2340 ± 50]*	[60	+14]											
P4	1 6	Ventersdorp lavas, Cape Province, 28 5S 25.0E	P€ [∼2100]*	211	+71	39	012½		+	55N	175E	130	22		S	
P5	1 7	Bushveld gabbro, Transvaal, 25 5S 28 0E	P€ [1950 ± 50]*	10	+59½	68	9½	$9.8^{+8.4}_{-4.5}$	+	23N	36E	41	12		N	1 07
P6	1 8	Waterberg sandstones, Transvaal, 23–26S, 28–30E	P€ [<1950 ± 50 / >1420 ± 70]*	see text						see text						
P7	1 9	Mashonaland dolerites, Rhodesia, 18 5S 31 5E	P€ [∼1600]*	112½	−51½	40	6½	$12\ 8^{+5.0}_{-3.6}$	M	7N	20W	28	7½		S	
P8	1 10	Premier Mine kimberlite, Transvaal, 25 5S 28 5E	P€	186	−24½	26	7½	$15\ 9^{+5.7}_{-4.2}$	−	51N	37½E	25	7½		S	
P9	1 11	Waterberg diabases, Botswana & Transvaal, 23–27S 25–30E	P€	189½	+ 2½	31	7½	$14.5^{+6.0}_{-4.2}$	M	65N	50½E	86	4½		S	
P10	1.12	Umkondo dolerites, Rhodesia, 19 0S 32 5E	P€	183	−10½	50	7½		−	65½N	40E	55	7		S	
P11	1 13	Umkondo lavas, Rhodesia, 20 5S 32 5E	P€	172½	−12½	13	15		−	61½N	15E	20	12		S	
	1 14	Umkondo lavas and dolerites (entries 1.12 and 1.13 combined)	P€	178	−11½	20	8	$18.1^{+6.1}_{-4.5}$	−	64N	26½E	27	7		S	
P12	1 15	Pilansberg dikes, Transvaal, 26 0S 28 0E	P€ [1310 ± 60]*	24	+69½	120	6	$7.4^{+6.4}_{-3.4}$	+	7½N	42½E	54	10½		N	1.06
	1 16	Fort Gouraund iron ore, Mauretania	P€													
	1.17	Van Dyk Mine dolerite dike, 26 2S 28 2E	P€ [≥1120]*	[158½	−66½	140	5]			[12½N	14E]			(7, 9)	S	

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TABLE 2. Paleomagnetic Results for the Phanerozoic of Africa

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										Paleomagnetic Pole						
Pole No.	Appendix Reference No.	Rock Unit and Location	Age, m.y.	D	I	k	α_{95}	Δ	Polarity	Latitude	Longitude	K	A_{95}	dp, dm	Pole Quoted	Reference *Irving* [1964]
	2 1	Lavas, east-central Morocco	€m [570 − 530]						M	[53N	34E]					
B1	2 2	Ntonya ring structure, Malawi, 15 4S 35.3E	P€-€ [~600]*	311½	+43	1054	2	$2\ 5^{+1.6}_{-1.0}$	+	27½N	15W	1130	2		N	
B1	2.3	Hook intrusives, Zambia, 15S 27E	€u-Ol [500 ± 17]*	300	+50	16	31½		+	14N	23½W	13	36		N	
	2.4	Dedza Mountain syenite, Malawi, 14.4S 34.3E		[308	+42]				+							
B3	2.5	Table Mountain Series, Cape Town, 33 1S 18.3E	S-Dl (430 − 390)	162	− 4		5		−	50N	11W			3, 5	S	4.09
	2 6	Table Mountain Series, Natal, 29.5S 30.8E	?S-Dl	[343	−56]				−	[74S	90E]	[94]	[6]		S	
	2 7	Dwyka varved clay, Rhodesia, 18S 29E	Cl (or Cu)	[0	−81	84	5½]		−	[36S	29E]			[10, 10]	S	6 56
	2 8	Dwyka varved clay, Rhodesia, 18S 29E	Cl (or Cu)	[333	+76	57	7]		+	[7N	17E]			[12, 13]	N	6.57
B4	2.9	Dwyka varves, central Africa, 9-18S 28-33E	Cl (or Cu)						M	26½S	26½E	52	10½		N	
B6	2.10	Ecca redbeds, Songwe-Kiwira, Ketewaka-Mchuchuma coalfields 9.4-10.7S 33.5-35.6E	Pl [280 − 260]	117	+54½	60	12		+	27S	89E	35	15½		N	
B5	2.11	K3 beds, Galula coalfield, 8 8S 32.9E	Cu or Pl	171½	+69	227	5		+	45½S	40E	91	8		N	
	2.12	Redbeds, Tanzania combined (entries 10, 11 combined)	?Cu-Pl						+	40S	64½E	11	16½		N	
	2.13	Ecca redbeds, Songwe-Kiwira coalfield, 9.5S 33.7E	Pl [280 − 260]	[156	+52]		[7]		+	[58S	74E]				N	
	2 14	Taru grit, Kenya, 3-4S 39-40E	'Pl'	[87	+61	23	16]		+	[0	87E]			[19, 25]	N	7.41
	2.15	Maji Ya Chumvi formation, Kenya, 3.1S 39.5E	'Pu'	[267	+38	9	11]		+	[4N	150E]			[8, 13]	S	7.40
	2 16	Microdiorite dike, Monkey Bay, Malawi, 14.1S 34.9E	Karroo	[63	+70]				+	[4N	67E]				N	
	2 17	Série rouge inférieure, Madagascar, 23 5S 44E	?P	[141	+45	13	4]		+	[53S	122E]			[3, 5]	N	
B7	2 18	Shawa ijolite, Rhodesia 19.2S 31.7E	Tr [209 ± 16]*	335½	−51½	46	11½	$11.9^{+10.2}_{5.4}$	−	64S	85½E	30	14		S	

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TABLE 2 (*continued*)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										Paleomagnetic Pole						
Pole No.	Appendix Reference No.	Rock Unit and Location	Age, m.y.	D	I	k	α_{95}	Δ	Polarity	Latitude	Longitude	K	A_{95}	dp, dm	Pole Quoted	Reference *Irving* [1964]
	2.19	Upper Beaufort sediments, S. Africa, 30S 28E	Trl						?	[67S	87E]				?	
	2.20	Red shale, Middle Beaufort, Natal, 28 7S, 29 2E	Trl	[36	−22	79	7]		−							
	2 21	Stormberg redbeds, O.F.S., S. Africa, 28.5S 28.6E	Tru	[7	−46	139	13½]		−							
	2 22	Sandstones, Rhodesia and Zambia, 16–20S 28.9E	Tru	[352	−31	100	7]		−	[84S	172E]			[4, 8]	S	8.27
	2 23	Red sandstone, Songwe rift valley, Tanzania, 9 5S 33E	Tr	[14	−27	23	10]		−							
B8	2.24	Red sandstone formation, Zambia, 16 2S 28 7E	Tru (200 − 180)	350½	−55½	216	4½	$5.5^{+4.0}_{-2.3}$	−	68S	49½E	178	5		S	
	2.25	Cave sandstone, Botswana, 23 5S 26.5E	Tru	[325	−13	7	8½]		−	[53S	136E]			[5, 9]	S	8.26
	2 26	Nuanetsi lavas, Rhodesia, 21.7S 31 3E	Tr (206 ± 13)*	[110	+72	8	25]		+	[29S	68E]			[39, 44]	N	
	2.27	Stormberg lavas, Lesotho, 29.6S 29 3E	Tru–Jl (200 − 150)	340	−49	9	14		M	72S	108E			13, 19	S	9.25
	2.28	Stormberg lavas, Sani Pass (normal), 29 6S 29 3E	Tru–Jl (200 − 150)	329	−54	66	4		−	64S	102E			3, 4	S	9.46
	2 29	Stormberg lavas, Sani Pass (reversed), 29 6S 29.3E	Tru–Jl (200 − 150)	162	+54	18	11		+	74S	96E			11, 16	N	9.48
	2.30	Stormberg lavas, Maseru (Normal), 29 4S 27.8E	Tru–Jl (200 − 150)	330	−50	40	5		−	64S	106E			5, 7	S	9.47
	2 31	Stormberg lavas, Maseru (reversed), 29 4S 27.8E	Tru–Jl (200 − 150)	178	+66	36	10		+	71S	33E			13, 16	N	9.49
B12	2 32	Stormberg lavas (entries 2.28–2 31 combined)	Tru–Jl (200 − 150)	338	−56	68		$9.8^{+10.5}_{-5.0}$	M	71S	89E	41	15		N/S	9.50
	2 33	Batoka basalts, Rhodesia, 18–20S 26–29E	Tru–Jl (200 − 150)	328	−40	187	5		−	60S	103E			3,5	S	9 24
	2.34	Karroo lavas, Rhodesia and Mozambique, 16–20S 28–34E	Tru–Jl [200 − 150]			incorporated in entry 2.35			−							
B13	2 35	Karroo lavas, central Africa (entries 2.33, 2.34 combined)	Tru–Jl [200 − 150]						−	57S	84E	41	8		S	

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TABLE 2 (*continued*)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										Paleomagnetic Pole						
Pole No.	Appendix Reference No.	Rock Unit and Location	Age, m.y.	D	I	k	α_{95}	Δ	Polarity	Latitude	Longitude	K	A_{95}	dp, dm	Pole Quoted	Reference *Irving* [1964]
	2.36	Karroo dolerite (surface collections) 30S 28E	Tru–Jl [190 − 154]*	352	−62		12		M	76S	52E			14, 19	S	9.26
	2.37	Karroo dolerite sill, Umgeni valley, 29 8S 31.0E	Tru–Jl	26	−67	286	7		−	62S	6W			10, 12	S	
	2.38	Limburgite plug, Rhodesia, 19.6S 29.9E	Tru–Jl	157	+50	101	5		+	66S	86E			5, 7	N	
	2.39	Karroo dolerites, Rhodesia, 20S 32E	Tru–Jl				incorporated in entry 2.48									
	2.40	Karroo dolerites, Winkelhaak, upper sill, 26.5S 29.1E	Tru–Jl	173	+58		5		+	77S	53E			6, 7	N	
	2.41	Karroo dolerites, Winkelhaak, lower sill, 26.5S 29.1E	Tru–Jl	307	−63		12		−	44S	82E			14, 18	S	
	2 42	Karroo dolerite, Estcourt (reversed), 29.1S 29.9E	Tru–Jl	167	+51		6		+	79S	101E			6, 8	N	
	2.43	Karroo dolerite, Estcourt (normal), 29.1S 29.9E	Tru–Jl	331	−64		8		−	62S	75E			10, 13	S	
	2.44	Karroo dolerites, mines and tunnels (entries 2.40–2.43 combined)	Tru–Jl						M	66S	78E	23	20		N/S	
	2.45	Baked Karroo sediments, Estcourt (normal), 29.1S 29.9E	Tru–Jl	324	−62		7		−	58S	84E				S	
	2.46	Baked Karroo sediments, Estcourt (reversed), 29.1S 29.9E	Tru–Jl	180	+55		11		+	84S	30E				N	
	2.47	**Karroo dolerites and baked sediments from mines and tunnels (entries 2.40–2.43 and 2.45, 2.46 combined)**	Tru–Jl	160	+60	52	9		M	68S	77E	25	14		N/S	9 27
B11	2.48	Karroo dolerites combined (entries 2.37–2.43 combined)	Tru–Jl						M	65½S	75E	16	12½		S	
B9	2.49	Marangudzi ring complex, 22.0S 30.7E	Tru [196 − 182]*	159½	+43½	41	8½	$12\ 6^{+7.2}_{-4.6}$	M	70S	105E	40	9		N	
B10	2.50	Mateke Hills ring complexes, 21.8S 31.2E	Jl [177 ± 4]*	149½	+58½	112	6½	$7\ 6^{+5.6}_{-3.1}$	+	58½S	79½E	66	8½		N	

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TABLE 2 (*continued*)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										Paleomagnetic Pole						
Pole No.	Appendix Reference No.	Rock Unit and Location	Age, m.y.	*D*	*I*	*k*	α_{95}	Δ	Polarity	Latitude	Longitude	*K*	A_{95}	*dp*, *dm*	Pole Quoted	Reference *Irving* [1964]
	2 51	Red siltstone, Mozambique, 16 2S 34 2E	?Kl	359	−46	160	6		–	79S	28E			5, 8	S	10.16
B15	2 52	Lupata series volcanics, Mozambique, 16.6S 34 2E	Kl [111 − 106]*	336	−54½	339	3		–	62S	79½E		4	3, 5	S	10.15
B14	2 53	Mlanje Massif syenite, Malawi, 16 0S 35 5E	Kl [116 − 128]*	333	−54	36	9½		–	60S	82E		12	9, 13	S	
	2 54	Lupata series and Mlanje Massif (2 52, 2 53) combined	Kl [106 − 128]*	334½	−54	66	4½	10 $0^{+3.6}_{-2.7}$	–	61S	80½E		6		S	
	2 55	Red sandstone, Songwe River Tanzania, 9 5S 33E	K	[21	−54	20	10½]		–	[58S	1E]			[10, 15]	S	
	2 56	Lavas and dikes, Madagascar	K [110 − 80]						–	68S	12E				S	10.08
	2 57	Lavas, Madagascar, 23 5S 44 3E	K [110 − 80]	44	−61				–	46S	2E				S	
	2 57	Lavas, Madagascar,														
B16	2 58	Ethiopian traps, 10.1N 38.3E	Ku-Te [∼69]*	358	+15	17	11½	19 $6^{+8.9}_{-6.1}$	–	87S	73½E	20	10½		S	
B17	2 59	Turkana lavas, Kenya, 2.9–4 3N, 34 4–35 8E	To-Tm [∼30]*	2	− 1	30	3	14 $8^{+3.4}_{-2.7}$	M	85S	13E			1½, 3	S	
	2 60	Cuirasse ferrugineuse, Madagascar, 22 5S, 45 4E	'Neogene'	[14	−37	27	8½]		–	[77S	53W]			[6, 10]	S	
B18	2 61	Lavas, Nairobi district, 1 3S 36 8E	[Tm]-Tp	7½	−5	46	10		M	82S	41W			5, 10	S	
	2 62	Rungwe volcanics, Tanzania, 9 0S 33 5E	Tp-Tpl	16	−28½	22	11½		M	73S	34W			7, 12	S	
	2 63	Rift Valley lavas, Kenya, 0.5S 36 4E	Tpl	192	+11½	40	7½		+	77S	29W			4, 7	N	
B19	2 64	Rungwe and Kenya Rift Valley lavas (2 62, 2 63 combined)	Tp-Tpl						M	74S	45W		10½		N	
	2 65	Lavas, south rift, Kenya, 1 0S 37.0E	Tp-Tpl	[359	+6½	9	5]		M	[85½S	155W]			[3, 5]	S	
	2 66	Bed I, Olduvai Gorge, Tanzania, 3 0S 35 3E	Tpl [1.9]*	2	−3				–	87½S	91½W				S	
	2 67	Aden volcanics, 12 8N 45.0E	Tp (∼5)*	173	−24	340	2½		M	83S	130E			1½, 3	N	12 42
	2 68	Ahaggar basalts, 23N 5E	Quaternary	345	+32	24	14		–	74S	76E			9, 16	S	12 44
	2 69	Archaeological baked clays, 37N, 10E	146 B.C.–300 A.D.	359	+54				–	88S	25E				S	12 43

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Columns 5 and 6 give the mean direction of magnetization specified by the declination D (degrees E of true N) and inclination I (measured from the horizontal and regarded as positive downward). For rock units in which the beds are undisturbed the paleohorizontal is the present-day horizontal, but if the beds have been tilted the paleohorizontal is assumed to have been the bedding plane, which in the case of sills and lavas is usually determined from adjacent sediments. If results contain directions of both positive and negative polarity (see column 10), the estimate of the mean direction (and of the precision of the result, columns 7, 8, 9, 13, 14, 15) is made irrespective of sign, and the direction quoted is the polarity of the majority of the results.

Columns 7 and 8 contain estimates of the precision (k) and the semiangle of the cone of 95% confidence (α_{95}) about the mean direction [*Fisher*, 1953]. These estimates and the similar data for poles in columns 13–15 are not always comparable between entries because they have been calculated using different weighting procedures according to the number of specimens, samples, or sites and according to variation in within-sample, within-site, and between-site precision. In each case the procedures are indicated in the notes.

Column 9 gives the angular dispersion $\Delta = 81/k^{1/2}$, which is a measure of the paleosecular variation. Ninety-five per cent confidence limits for Δ are given according to the method outlined by *Gough et al.* [1964*b*]. These limits are not the same on either side of the value of Δ because of the procedure involved in their calculation. The angular dispersion is given only for those studies which can warrant it, that is, where sampling is believed to have been adequate for a full record of paleosecular variation to have been obtained.

Column 10 gives the polarity. For Tertiary and Recent time this is simply defined so that directions in the same sense as the present earth's field are called negative (normal) and antiparallel directions are called positive (reversed). This convention works well for as far back as the Permian, but its rigid application would necessitate the drawing of a 180° shift in the polar-wander curve (Figures 1 and 4) where that curve crosses the present equator. Such a shift would not reflect an actual geophysical event (reversal of the field). Therefore, for Precambrian and Paleozoic rocks, the poles are plotted to give continuity to the polar-wander curves. For the Precambrian this convention is completely arbitrary and will remain so until the gaps in the polar-wander curves can be filled. Polarity is now defined with respect to the poles on polar-wander curves, as drawn in Figures 1*a* and 4. This convention provides continuity with the definition given above for Tertiary and Recent rocks. The occurrence of mixed polarities is denoted by M.

Columns 11 and 12 give the latitude and longitude of the paleomagnetic pole. The pole which is quoted is the one on or near the polar-wander curves in Figures 1*a* and 4, rather than its antipole.

Columns 13 and 14. If the pole was calculated by applying Fisher statistics to poles obtained at a lower sampling level [*Creer et al.*, 1959], then precision

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estimate K and semiangle of the cone of 95% confidence A_{95} about the mean pole are given.

Column 15 gives the semiaxes of the oval of confidence about the paleomagnetic pole at the 95% probability level. *dp, dm* are, respectively, along and perpendicular to the paleomeridian through the pole and sampling site [*Irving*, 1956]; these parameters are quoted if the pole was calculated directly from the mean direction and the Fisherian precision.

Column 16. The pole quoted in columns 11 and 12 is designated 'north' or 'south.' Where mixed polarity occurs, the designation corresponds to the polarity of the majority of the results.

Column 17 gives the reference numbers in the appendix of *Irving* [1964] where applicable. Entries which were discussed by Irving are not discussed at length here, and reference should be made to the earlier discussion. Precision estimates are quoted to the nearest whole number, and angles to the nearest half-degree, except for some poles and for Δ, for which more precise definition is justified by the statistics.

PRECAMBRIAN

African polar-wander curve. The paleomagnetic poles listed in Table 1 apply to Africa alone. They may not apply to the whole of Africa if there have been large relative movements within that continent. The possibility that large relative movements occurred after the magnetization of a particular rock unit necessarily increases with its age. The use of single Precambrian poles to draw paleolatitude lines over the whole of Africa is certainly not justified. The results given here (Table 1) all come from the Transvaal and Rhodesian shields and should not be taken to apply to other parts of Africa.

Figure 1 shows the paleomagnetic poles relative to the Transvaal and Rhodesian shields. Two polar-wander curves have been drawn through these representing two different time intervals. The first curve (Figure 1*a*) covers the interval 2700 to about 2100 m.y. (P1–P4). To draw these curves one starts with the pole for the oldest rock unit (P1), and in Figure 1*a* the pole which occurs nearest to the sampling locality has been plotted. Pole P1 is then joined to P2 through the shortest possible path. There are two paths, either the one shown in Figure 1*a* from P1 to P2 or the one from P1 to the antipole of P2 (P2′ in Figure 1*b*). It is clearly more realistic to postulate the minimum amount of polar wandering for any given time interval; therefore we adopted the path shown in Figure 1*a*. The procedure is then repeated from P2 to P3 and so on. The continuity of the polar path from P1 to P4 supports the ages suggested by radioactive age determinations [*Evans*, 1967]. There is a possibility of large relative movements between the Transvaal and Rhodesian shields before about 2000 m.y.; it may be that the pole for the Great Dyke of Rhodesia (P2) should not be fitted to the curve drawn through the three poles for the Transvaal shield for the time before 2000 m.y. (P1, 3, 4).

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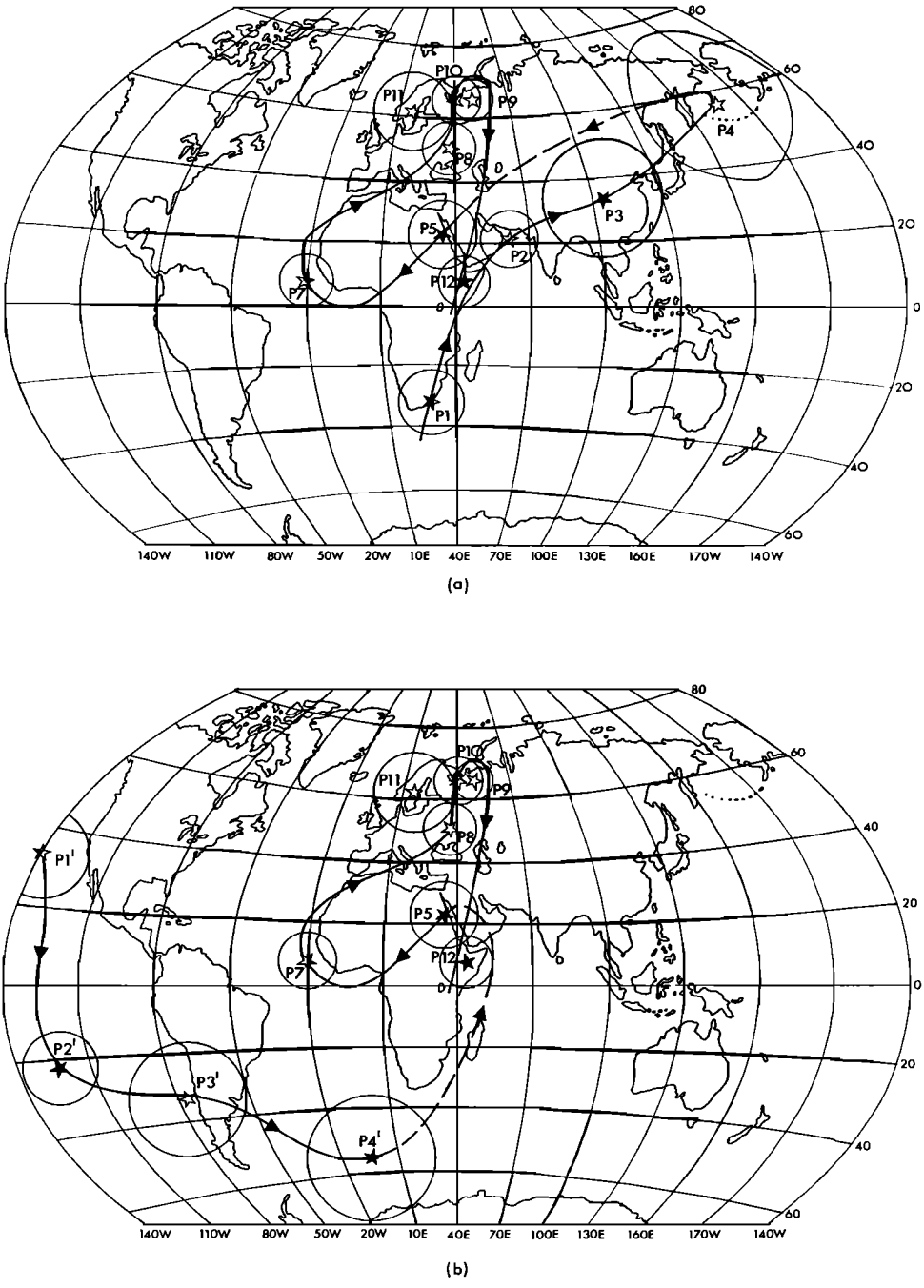


Fig. 1. Precambrian polar-wander curve for Africa (*a*) using the poles as listed in Table 1 and (*b*) using the antipoles of P1 to P4 as explained in the text. Solid symbols are north poles, open symbols south poles. Where both polarities occur, the proportion of north and south polarities is indicated. Poles are numbered as in Table 1, and 95% confidence limits are shown.

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Figure 1 also shows the polar-wander curve relative to the Transvaal and Rhodesian shields for the interval 2000 to 1300 m.y. The two shields are considered a unit during this period because radioactive age determinations indicate that the close of major orogenic events in the intervening Limpopo belt was about 2000 m.y. ago [*van Breemen et al.*, 1966]. Two series of results are summarized by this curve—those from a variety of basic igneous rocks which have been dated radiometrically (P5, 7–12) and those from the sediments of the Waterberg system. These sediments (P6) do not show a single paleomagnetic pole covering the time of deposition of the redbeds. Instead, a polar-wander path covering the period of deposition and consolidation of the sediments is reflected in the paleomagnetic results. *Jones and McElhinny* [1967] have shown that the polar-wander paths for the sediments and the igneous rocks can be superimposed, and the path drawn in Figure 1 follows the one they have derived.

The manner in which the older and younger parts of the curves in Figure 1*a* should be joined is not at present clear. The gap between about 2100 m.y. (P4) and 1950 m.y. (P5) is represented by a great-circle distance of about 100°. This gap would be reduced to 80° if the antipoles were used for the curve for either set. When the antipoles of the older curve are taken (P1′–P4′), the resulting polar-wander curve is that shown in Figure 1*b*.

Between 2700 and 2100 m.y. about 150° of polar wandering appears to have taken place at what appears to be a fairly uniform rate of 0.25° per million years. Between 1950 and 1300 m.y. about 180° of polar wandering seems to have taken place, so that the average rate does not differ significantly from that deduced from the older part of the curve. It is unlikely that this wandering took place at a uniform rate, however, since 120° of the movement appears to have occurred during the deposition of the Waterberg system which may represent only 100 m.y. This large amount of polar wandering within the time span of the Waterberg system produced considerable latitude change (Figure 2). It was not, however, accompanied by great changes of sedimentary facies, which are largely redbeds throughout. Redbeds also occur in intermediate latitudes in the Karroo system; these two examples call into question the (qualified) conclusion of *Briden and Irving* [1964] that redbeds occurred preferentially in low paleolatitudes, a somewhat surprising conclusion in view of the known variety of origins of redbeds [*van Houten*, 1961].

In Figure 2 paleolatitudes have been drawn for various stages of the Precambrian. Latitude changes from near polar (2700 m.y.) to near equatorial (2350 m.y. and upper Waterberg) are suggested. The paleolatitudes are drawn only across the areas to which they may properly be applied, and their extension across adjacent cratons is dotted.

Paleomagnetic correlation and dating. Jones and McElhinny [1966] have demonstrated that paleomagnetism can be a powerful tool for geological correlation, especially across large distances of Precambrian shield areas. Figure 1 has already demonstrated that polar movement has been substantial throughout the Precambrian. Thus, if two rock units can be shown to have the same paleomagnetic pole, they are likely to be approximately the same age.

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Fig. 2. Precambrian paleolatitudes for southern Africa.

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When sufficient data are available it should be possible to determine the age of a rock unit by noting where its paleomagnetic pole falls on the polar-wander curve, thus providing an alternative or supplementary method of dating to radioisotopic age determinations. For example, the only previous evidence for the age limits of the Waterberg system was that it overlies sediments which have variously been regarded as pre-Waterberg ('Loskop system') or lower Waterberg. These sediments are intruded by a granophyre which has been dated at 1790 ± 70 m.y., and they overlie the Bushveld igneous complex (1950 ± 50 m.y.). Also, the Waterberg system is intruded by the Leeuwfontein complex, dated at 1420 ± 70 m.y. More precise radiometric dating of the Mashonaland dolerites and Umkondo dolerites of Rhodesia, combined with the superimposition of the polar-wander paths from the Waterberg sediments and from these dated igneous rocks, will make it possible to determine the age of the Waterberg system more exactly.

The 'pole' from the Fort Gouraud iron-ore body of Mauretania, North Africa (entry 1.16 in Appendix 1), lies within the circle of confidence of the pole from the Mashonaland dolerites. *Porath* [1967] suggests that they might therefore be contemporaneous and that this agreement serves to date the formation of this ore body. Such agreement should be treated with caution in view of the possibility of relative movements between the Rhodesian and West African cratons.

Intercontinental correlations, Africa and Australia. Porath [1967] compared the paleomagnetic poles derived from Australian Precambrian iron ores with the younger part of the African curve (Figure 1). After placing Africa and Australia in their relative positions according to *Carey*'s [1958] reconstruction of Gondwanaland, Porath then attempted to date the ore bodies from the positions of their poles on the African curve. Unfortunately, the relative ages of the Australian ore bodies are not known, but they are thought to lie within the range of the younger part of the African curve (1950 to 1300 m.y.). When the reconstruction is made, the Australian poles group quite well around the region of the African curve. This general agreement might be taken as an indication that Africa and Australia have remained in the same relative position from the middle Precambrian to the time of the breakup of Gondwanaland. Until more information is available from the Australian Precambrian, no firm conclusions can be drawn.

Intercontinental correlations, Southern African and Canadian shields. It is only from the Canadian shield that there are sufficient paleomagnetic data to permit us to attempt a comparison with the Southern African Precambrian results. Table 3 summarizes the results from diabase dikes of the Canadian shield dating from about 2000 to 1200 m.y. The poles for the Canadian results are plotted in Figure 3*a*, and a polar-wander curve *can* be drawn through them of roughly the same shape and size as the younger African curve. This is demonstrated in Figure 3*a*, where the African curve has been superimposed upon the Canadian poles.

Many of the Canadian data are of a preliminary nature, and, except for the Abitibi and Sudbury swarms, samples were cleaned in alternating fields of up to only 130 oe. The dispersions are calculated by giving samples unit weight and should be treated with caution, especially for the larger collections. The K-Ar age determinations must be regarded as tentative in most cases because of the small

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TABLE 3. Paleomagnetic Results and K–Ar Ages of Diabase Dikes from the Canadian Shield

	Dike Swarm	Reference	D	I	N	R	k	α_{95}	Paleomagnetic Pole			K–Ar Ages, Sample Numbers, and References
									Latitude	Longitude	dp, dm	
7.	Mackenzie	*Fahrig et al.* [1965]	251	+25	34 samples	29.1	7	10	4N	177W	6, 11	1295 m.y., average of 4 biotites [*Fahrig and Wanless*, 1963] 1390 m.y., 1 whole rock [*Fahrig et al.*, 1965]
6.	Sudbury[1]	*Larochelle* [1967]	268	−1	53 samples	52 0	52	3	2S	171W	1, 3	1020 m.y. [*Fairbairn et al.*, 1960] 1240 m.y. (GSC 64–100) [*Wanless et al.*, 1966] 1330 m.y. (GSC 63–117) [*Wanless et al.*, 1965]
5.	Abitibi ENE dikes, group 1	*Larochelle* [1966]	264	+61	10 sites	9 91	102	5	27N	134W	6, 7	1240 m.y., biotite (GSC 63–118) 1220 m.y., biotite (GSC 63–130) 1825 m.y., whole rock (GSC 63–149) [*Wanless et al*, 1965]
4.	Molson	*Fahrig et al.* [1965]	210	+80	5 samples	4.8	27	15	36N	109W	27, 28	1445 m.y., 1 whole rock (GSC 64–77) [*Wanless et al.*, 1966]
3.	Abitibi ENE dikes, group 2	*Larochelle* [1966]	354	−32	2 sites	1.97			24N	107E		
2.	Abitibi NNE dikes	*Larochelle* [1966]	221	+19	1 site				21S	122W		
1.	Marathon[2]	*Fahrig et al.* [1965]	268	+60	9 samples	8.7	25	11	29N	147W	12, 16	1810 m.y., 1 whole rock (GSC 63–123) [*Wanless et al.*, 1965]

[1] Also 1205 ± 50 m.y. Rb–Sr [*van Schmus*, 1964].
[2] 'Two anomalous samples excluded from average.'

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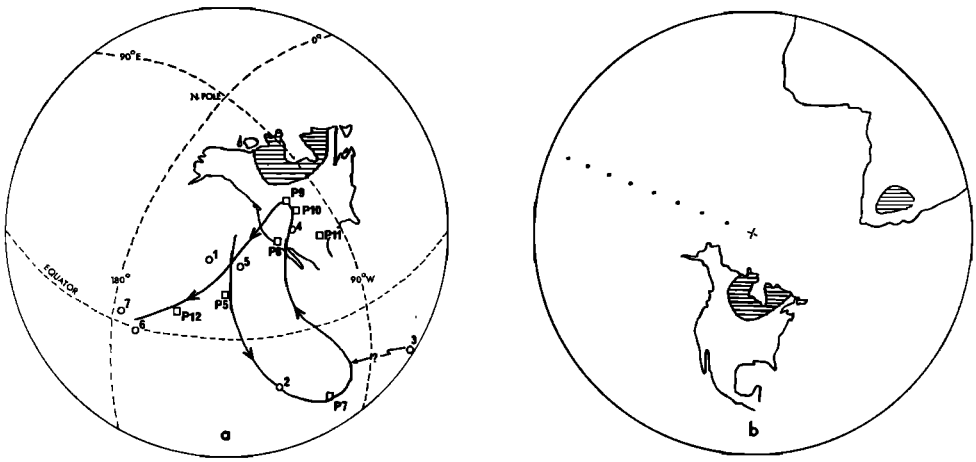


Fig. 3. (*a*) African polar-wander curve of Figure 1 superimposed upon the Canadian poles listed in Table 3. Squares, African poles; circles, Canadian poles. (*b*) Corresponding relative positions of Southern African and Canadian shields.

number of samples involved in these determinations and because of the complexities of interpreting isotopic age data from the Canadian shield. Nevertheless, it may be justifiable to infer that the similarity of the African curve to the positions of the Canadian poles confirms the amount of polar wandering during the time interval which they cover. The position of the African curve in relation to the Canadian poles in Figure 3*a* fixes the relative positions of the Southern African and Canadian shields during this part of the Precambrian (Figure 3*b*). We therefore conclude that the Southern African and Canadian shields did not move relative to each other by a detectable amount between 2000 and 1300 m.y. and that the polar-wander curve reflects *either* the joint drift of the Southern African and Canadian shields *or* polar wandering *or* a combination of the two.

This is the first time that two Precambrian polar-wander curves have been successfully superimposed, and, if the procedure is correct, the correlation of the Southern African and Canadian rocks which have been studied paleomagnetically can be deduced as in Table 4. In compiling this table we considered paleomagnetic correlation to overide the more tentative age determinations; this is a perfectly legitimate decision with the data as they stand, but the correlations will have to be tested by further work.

PHANEROZOIC

Table 2 summarizes all African Phanerozoic results to date. Of the 69 entries only 19 are considered reliable. The reasons for not using the rest of the data are given in Appendix 2.

Polar-wander curve for the Paleozoic. The oldest rocks studied are about 600 m.y. old (B1), and, because these results are the best thus far obtained for the lower Paleozoic, the curve in Figure 4 is drawn from this pole to B4. The glacial varves of central Africa (B4) are commonly regarded as being Late Carboniferous

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TABLE 4. Tentative Correlation of Formations Studied Paleomagnetically from Southern African and Canadian Shields

	Southern Africa	Canada
		Sudbury and Mackenzie dikes
	Pilansberg dikes (1310 ± 60 m.y.)	
		Abitibi ENE dikes, group 1
↑	Umkondo and Waterberg intrusives	
	Umkondo lavas (≥1420 ± 70 m.y.)	Molson dikes
Waterberg system	Premier Mine kimberlite	
		Abitibi ENE dikes, group 2
↓	Mashonaland dolerite (≥1600 m.y.)	
		Abitibi NNE dike
	Bushveld igneous complex (1950 ± 50 m.y.)	
		Marathon dikes
	Ventersdorp lavas (~2100 m.y.)	

in age, though on paleontological evidence they can only be said to be 'probably Carboniferous.' *McElhinny and Opdyke* [1968] argue on the basis of a comparison with paleomagnetic results for the Carboniferous of Australia [*Irving*, 1966] that the varves are in fact Lower Carboniferous. E. P. Plumstead (private communication), however, now regards the lowest Dwyka glaciation in South Africa as being Lower Carboniferous or even Upper Devonian, because of evidence from plant fossils lying directly upon a tillite in the Orange Free State. The angular difference between poles B1 and B4 is about 60°, and this polar movement must have occurred at unknown time(s) between about 600 and 330 m.y.; the average rate is similar to that calculated for both parts of the Precambrian discussed earlier. But the result from the Table Mountain Series (B3) suggests that all this drift may have taken place after Late Silurian time, which would imply a much higher drift rate, and *Briden* [1967*b*] has suggested, after comparison of African, South American, and Australian evidence, that it all occurred during Middle Devonian time.

The K3 redbeds of southern Tanzania are regarded as Lower Permian, but they yield poles B5 and B6, which are significantly different from one another (Figure 4). *Opdyke* [1964*b*] noted this difference but was undecided whether it was due to inadequate sampling or to an error in correlation between the different coalfields which were sampled. On the latter interpretation the difference between poles could be due to polar wandering. The evidence for Early Permian age is firmer for pole B6 than for B5. *Opdyke* [1964*b*] mentions the possibility that the Galula redbeds (pole B5) might be slightly younger than the others, but only because it is not possible to set an upper limit to their age, whereas the Ketewaka-Mchuchuma beds are stated to be not younger than Middle Artinskian [*Hart*, 1960]. The possibility that the Galula K3 beds might be older than the others did not occur to *Opdyke* [1964*b*] when he first reported the paleomagnetic measurements. However, *McElhinny and Opdyke* [1968] suggest that this is indeed the case and that the polar movement follows the sequence B4–B5–B6. They point

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out that this makes the Permo-Carboniferous results for Africa and Australia consistent with *King*'s [1962] reconstruction of Gondwanaland. On this view the polar-wander curve takes the solid line in Figure 4, but, if the difference between poles B5 and B6 occurs because sampling has not separately averaged out the paleosecular variation, they should be combined, as was done by *Opdyke* [1964*b*]. The polar-wander curve then follows the dashed line in Figure 4, from B4 to (B5 + B6) to the Mesozoic poles around 65°S. This alternative has also been discussed by *Briden* [1967*b*], who, on the basis of comparison of the African Australian, and South American Permo-Carboniferous results, points out that it implies that *all* the K3 beds from Tanzania are most likely to be Upper Carboniferous and that the combined pole (B5 + B6) refers to this period. Both of these possibilities have to be raised because they result in different interpretations of the movement and fragmentation of Gondwanaland.

Polar-wander curve for the Mesozoic to the present. There are now nine poles from extensive studies of a wide variety of igneous rocks and sediments of Triassic to Early Cretaceous age, and they all fall in a group around 65°S, 80°E. It may be that the statistically significant differences between some of them are due to inadequacies of sampling or small undetected tectonic disturbances. On this view all these poles could be combined ($N = 9$, $R = 8.9283$, $K = 111$, Lat. = 64.5°S, Long. = 80.8°E, $A_{95} = 4.8°$), and it would be concluded that Africa was stationary relative to the pole from the beginning of Mesozoic time until about 100 m.y. ago (designated a quasi-static interval by *Briden* [1967*b*]). The results, if taken at their face value, however, seem to indicate small movements (≈10°) on more than one occasion during Late Triassic–Early Jurassic time [*Brock,* 1968], but the close grouping of the poles from the Mateke Hills complexes (B10), the Lupata volcanics and the Mlanje Massif (B14, B15) strongly suggests that Africa remained fixed from 180 to 100 m.y.

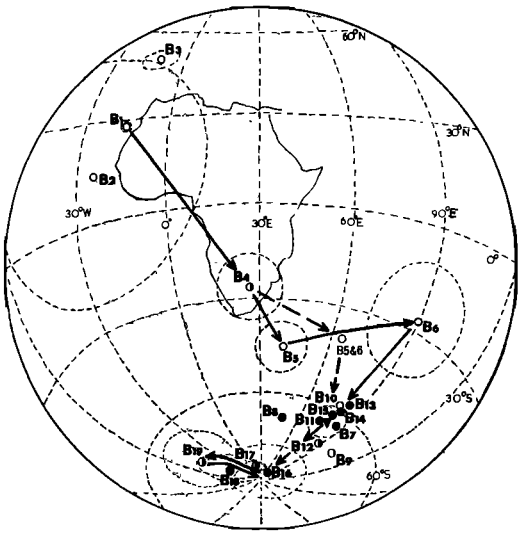


Fig. 4. Phanerozoic polar-wander curve for Africa. Poles are numbered as in Table 2. Solid symbols, south poles; open symbols, north poles. Where both polarities occur the proportion of north and south polarities is indicated. Circles of 95% confidence are shown for the poles. The triangle and its circle refers to the mean Mesozoic pole (B7 to B15 inclusive).

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Pole B16 for the Ethiopian lavas of Eocene age lies very close to the present pole and suggests a shift of 20° in the interval 100 to 70 m.y. ago. This corresponds to an average rate of polar movement nearly twice that observed for Paleozoic and Precambrian times. Three results from Tertiary lavas in East Africa suggest the south pole may have made a westward excursion of about 15°, mainly during the late part of the Tertiary.

Paleolatitudes and paleoclimatic indicators. A set of paleolatitude maps for Africa can be drawn by using the paleomagnetic poles (Figure 5). There is less probability that relative movement occurred within the continent in Phanerozoic time than in Precambrian time, and consequently the paleolatitude lines can be extended farther from the sampling areas with more confidence than was done in Figure 2.

The paleolatitude maps of Figure 5 can be assessed in terms of paleoclimatic data. If the paleomagnetic latitudes are consistent with the paleoclimatic evidence, then this agreement not only supports the validity of the paleomagnetic results but also provides confirmation that the poles are both paleomagnetic and

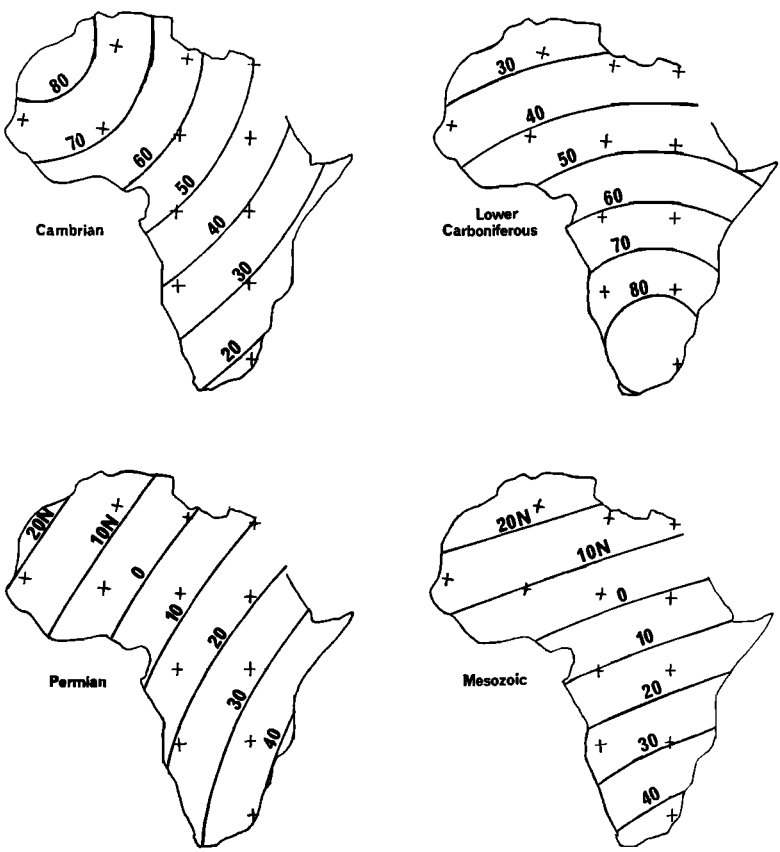


Fig. 5. Phanerozoic paleolatitudes for Africa. Latitude lines have been drawn using poles B1, B4, B6, and the mean Mesozoic pole (B7–B15).

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paleogeographic. Such a comparison is, in other words, a test of the axial property of a geocentric dipole field. A proper discussion of the distribution of paleoclimatic indicators requires a full series of lithofacies maps and faunal and floral distribution maps. These do not exist, and all that we attempt here is to point out the paleolatitudes of some well-known stratigraphic features.

The data indicate that polar shifts relative to Africa in Tertiary time amounted to no more than about 15°. Because the shifts have been roughly at right angles to the paleomeridians through Africa, they involve very little latitude change. Therefore the distribution of paleoclimatic indicators is likely to be compatible with the present framework as well as with Tertiary paleolatitudes. In discussing Mesozoic time it is sufficient to assume here that the pole was stationary round 65°S, 80°E, throughout. This places Natal closest to the pole at about 40°S, which is consistent with the occurrence in the Mesozoic sequence of limestones and faunas adapted to a warm climate. The Permian results place North Africa and Arabia in low paleolatitudes, and this is consistent with a predominantly carbonate sequence in the marine Permian of Tunisia and with the occurrence of evaporites in the Permian of Arabia.

The great Carboniferous glaciation in southern Africa has been well documented, and the paleomagnetic pole at that time was situated in South Africa itself. *Du Toit* [1954] has concluded that the glacial deposits on the western side of Africa are generally older than those to the south and east and that the centers of glaciation moved from west to east, roughly along the tropic of Capricorn. This was part of a general west-east trend of centers of glaciation, the oldest being in South America and the youngest in Australia. This interpretation is in line with the paleomagnetic evidence that the pole moved southeastward from the Congo and then more eastward from southern Africa (Figure 4).

Poles B1 and B2 for the Cambro-Ordovician period place North Africa in high paleolatitudes, the pole lying just off the Mauretanian coast. Evidence for glacial conditions in the Sahara at this time is now well established, and it appears that a huge icecap covered this part of Africa at that time [*Beuf et al.*, 1966]. The only recognizable marine rocks in Africa south of the Sahara which might belong to this period are the stromatolitic limestones of the Nama system (Southwest Africa); these would be placed in low paleolatitude, which is consistent with their supposed affinity for a warm environment.

COMPARISON OF THE AFRICAN PHANEROZOIC RESULTS WITH THOSE FOR THE OTHER SOUTHERN CONTINENTS

Comparison of the African Phanerozoic data with those for the other southern continents will be confined to the major points which arise. In this review of *African* data it is not our intention to make a detailed analysis of all Phanerozoic data for the southern continents, since this has been undertaken elsewhere [*Briden*, 1967*b*; *McElhinny*, 1967].

A single pole from a continent gives information on its latitude and azimuthal orientation but not on its longitude. Information on *relative* longitudes of two continents can be obtained, however, by examining paleomagnetic data for successive geological periods. As *Irving* [1958] and *Graham et al.* [1964]

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have pointed out, the relative positions of two continents can be determined uniquely if they remained in a fixed relation during a period of substantial polar wandering.

Cambrian to Carboniferous results for Africa and South America. For the lower Paleozoic there are at present reliable results only for Africa (Table 2) and South America [*Creer*, 1965]. The Cambrian and Ordovician (B1, B2), Silurian (B3), and (Lower?) Carboniferous (B4) poles for Africa are compared in Figure 6*a* with corresponding poles for South America. When the South Atlantic Ocean is closed and Africa and South America joined as in Figure 6a, the agreement between these poles is remarkable. This comparison confirms the hypothesis that the South Atlantic was closed from the Cambrian until at least the Carboniferous. The relative longitudes are not so well determined for the time since the Carboniferous, but the relative positions in Figure 6*a* are similar to those in the Mesozoic reconstruction (Figure 7*b*), and it may be that the opening of the South Atlantic did not begin until the Cretaceous.

Upper Paleozoic results for Africa and Australia. Polar movement relative to Australia during the late Paleozoic has been documented by *Irving* [1966]. *McElhinny and Opdyke* [1968] have inferred similar movement relative to Africa corresponding to the sequence of poles B4–B5–B6 (Figure 4). If their interpretation is correct, it leads to the relative positions of Africa and Australia

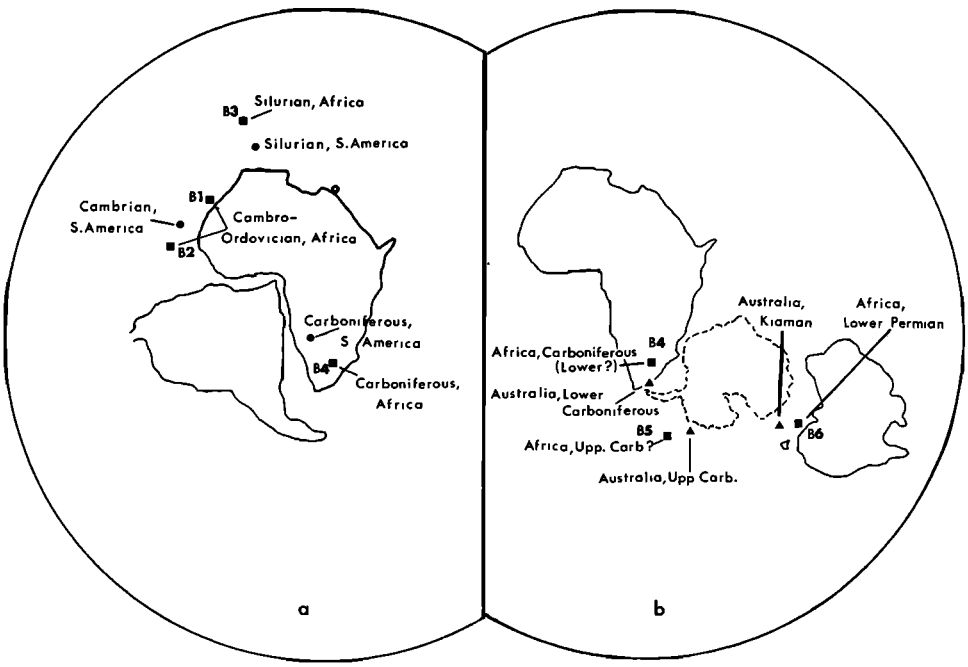


Fig. 6. (*a*) Relative positions of Africa and South America obtained by bringing the Paleozoic poles into coincidence. (*b*) Relative positions of Africa and Australia obtained by bringing the upper Paleozoic poles into coincidence. The position of Antarctica is that shown in Figure 7.

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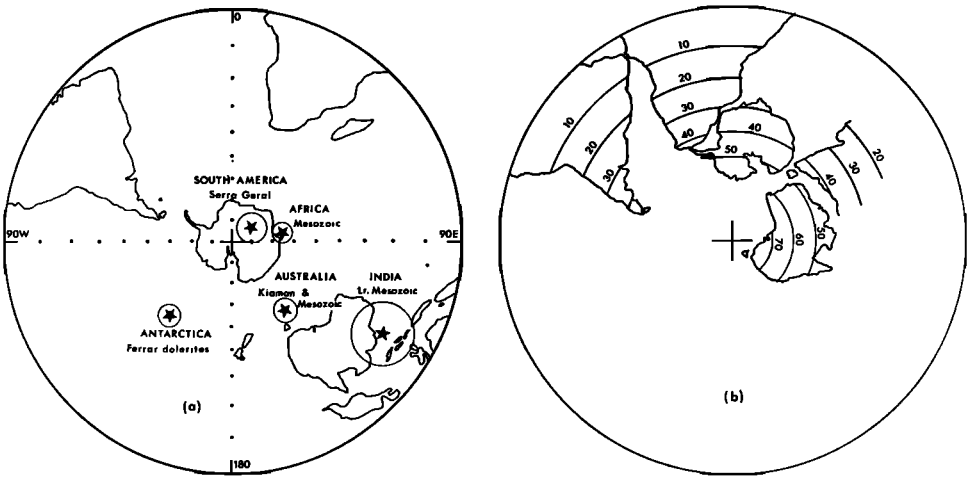


Fig. 7. (*a*) Mesozoic poles and their circles of 95% confidence for the southern continents. (*b*) Reconstruction of Gondwanaland obtained by bringing the poles into coincidence. Paleolatitude lines are drawn across the continents.

during the late Paleozoic shown in Figure 6*b*. The position of Antarctica is dashed and corresponds to its position relative to Africa in Mesozoic time (Figure 7*b*). This part of the reconstruction is very close to that suggested by *King* [1962] for the Upper Carboniferous.

Figures 6*a* and 6*b* can be combined to form a reconstruction of the southern continents for the Lower Carboniferous. The reconstruction could be valid for the whole of Paleozoic time because, in addition to the data shown in Figure 6, it is compatible with Devonian data for South America and is not seriously contradicted by tentative results for the Cambrian of Australia. This reconstruction is different in detail from the Mesozoic reconstruction (Figure 7*b*) which is discussed next, however, and the timing of the rearrangement of the configuration is a matter for debate.

Mesozoic results for Gondwanaland. It has already been pointed out that during Mesozoic time the pole relative to Africa appears to have remained fixed. A similar situation exists for Australia from the uppermost Carboniferous to the middle of the Cretaceous. There are fifteen poles for Australia covering this period, and the mean of these is termed by *Irving* [1966] the 'mean Kiaman and Mesozoic pole for Australia,' which lies at 48°S, 142°E, with a 95% polar error of 5°. The African and Australian mean poles and their errors are plotted in Figure 7*a*.

Creer [1962*a*] has made a very detailed study of the Serra Geral formation in South America, whose age lies in the range Late Jurassic to Early Cretaceous [*Amáral et al.*, 1966; *McDougall and Rüegg*, 1966]. *Bull et al.* [1962] have made a similarly detailed study of the Ferrar dolerites of Antarctica, which are of Jurassic age [*McDougall*, 1963].

There is evidence from India that the pole may have remained stationary from Permo-Triassic to Middle Jurassic time. There are three poles in this

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interval, all derived from studies in which magnetic cleaning techniques were used, and *McElhinny* [1967] has averaged them to form the *Mean Lower Mesozoic Pole for India,* which lies at 16°S, 121°E, with A_{95}= 10°. This mean pole and the South American and Antarctic poles are also plotted in Figure 7*a*.

The poles for the five continents are widely divergent, a fact which has often been pointed out. The new African and Australian results, however, serve to confirm the divergence, especially between these two continents, and to demonstrate that there must have been relative movement between the southern continents since the Mesozoic era. In Figure 7*b* a reconstruction for the Mesozoic has been attempted using the data of Figure 7*a*. It has been made by assembling the closest grouping of the continents that is consistent with the paleomagnetic data. This reconstruction is almost identical with that suggested by *Gough et al.* [1964*b*] and bears some similarity to those suggested by *Du Toit* [1937] and *King* [1962]. Because of the absence of polar wandering during Mesozoic time, there is no direct paleomagnetic confirmation of the correctness of the relative longitudes shown in Figure 7*b*. However, the new African paleomagnetic data support this reconstruction indirectly, to the extent that the relative positions shown in Figures 6*b* and 7*b* are not grossly different. The detailed differences between these two reconstructions can be taken to imply relative movement between the continents during Late Permian time, if the African data are interpreted as has been done in Figure 6.

An alternative suggestion [*Briden,* 1967*b*] is that the late Paleozoic polar movement relative to Africa follows the sequence B4 to (B5 + B6) to the Mesozoic poles as was shown in Figure 4. This interpretation leads to relative movement between Africa and Australia occurring during the Carboniferous rather than the Permian but does not alter the general conclusion that Africa and Australia were in the relative positions shown in Figure 7*b* during the Mesozoic.

Jurassic to lower Tertiary results for India and Australia. Finally, the position of India in the reconstruction of Gondwanaland for the Mesozoic can be obtained by a comparison [*Irving,* 1964] of the Jurassic and early Tertiary poles of India and Australia.

In this comparison the more recent results (with magnetic cleaning) from the Rajmahal and Deccan traps, summarized by *McElhinny* [1967], have been used. The agreement between the pole pairs is then slightly improved in comparison with that shown by *Irving* [1964]. *McElhinny* [1967] has pointed out that the Deccan traps may be slightly older than the Tertiary basalts of New South Wales and Victoria, but, in spite of this, the comparison made in Figure 8 still serves to demonstrate the general correctness of the position of India alongside Australia in the Mesozoic reconstruction of Figure 7*b*.

PALEOSECULAR VARIATION

Creer [1962*b*] has analyzed the dispersion of the present geomagnetic field for comparison with paleomagnetic results. It is supposed that the time variations of the directions of the geomagnetic field at a given locality are comparable to the variations at a given epoch around a line of latitude passing

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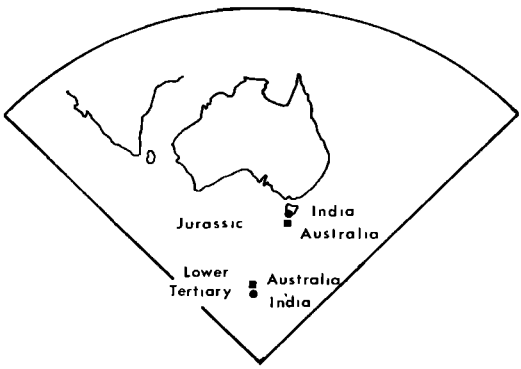


Fig. 8. Relative positions of India and Australia obtained by bringing the Jurassic and lower Tertiary poles into coincidence.

through that locality. The scatter of directions in the field variations around each line of latitude can be expressed in terms of the angular dispersion $\Delta = 81/k^{1/2}$ degrees, where k is the *Fisher* [1953] precision for these directions. The angular dispersion so calculated varies with latitude, the value at the equator (18.5°) being about twice that at the poles. This is consistent with Model A of *Irving and Ward* [1964], who suppose that a geocentric dipole field is perturbed by randomly directed components of constant magnitude. Paleomagnetic results provide estimates of the angular dispersion in the past (Δ_{paleo}) from the estimates k of the precision of directions in each rock unit. A comparison between present and past estimates of secular variation is relevant to the problem as to whether the earth's magnetic field has always been that of a geocentric dipole. Conformity between these estimates strongly suggests that this has been the case.

In Tables 1 and 2 values of Δ_{paleo} are given for those studies detailed enough to suggest that the paleosecular variation may be fully represented in the measurements. These are plotted in Figure 9 versus paleolatitude, and it is noted that only 3 of the 19 estimates are significantly different from the variation associated with the present earth's field (Δ_{pres}). In Figure 10 the values of Δ_{paleo} have been normalized to Δ_{pres} according to *Irving and Ward*'s [1964] Model A using the relation

$$\Delta_{pres} = 18.5(1 + 3 \sin^2 \lambda)^{-1/2} \quad \text{degrees}$$

where λ is the latitude equal to the paleolatitude at which Δ_{paleo} was determined and Δ_{pres} has the value 18.5° at the equator. In this way the paleosecular variation can be plotted as a function of time. None of the 9 Precambrian estimates and only 3 of the 10 Phanerozoic estimates differ significantly from 1.00.

APPENDIX 1. NOTES ON TABLE 1

1.1. *Modipe gabbro, Botswana* [*Evans and McElhinny,* 1966]. A result from $N = 10$ sites (36 samples from 18 cores and 3 hand samples) after magnetic cleaning of all samples in fields of 100 to 1000 oe. Thermal demagnetization experiments confirm the presence of a stable component. The gabbro has been intruded by the Gagerones granite (2340 m.y.). Thermal and magnetic cleaning

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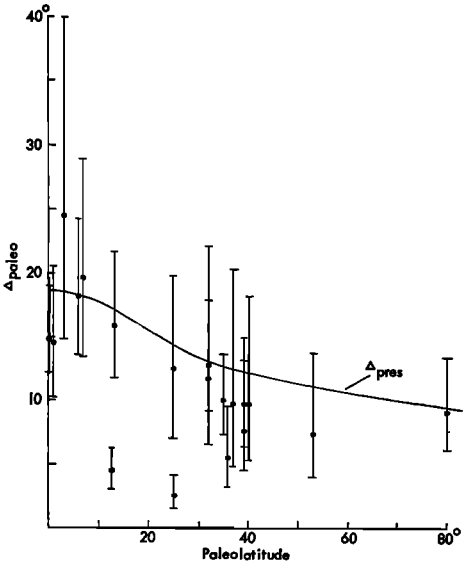


Fig. 9. Angular dispersion as a function of paleolatitude using the values shown in Tables 1 and 2. The curve represents the dispersion estimated for the present geomagnetic field. Ninety-five per cent confidence limits are shown.

of gabbro samples from the baked-contact zone confirm the stability of unbaked samples. Three sites were reversed with respect to the rest. The age was determined by *McElhinny* [1966*a*] using the Rb-Sr method on total rock and a decay constant of 1.39×10^{-11} yr^{-1}.

1.2 *Great Dyke, Rhodesia* [*McElhinny and Gough,* 1963]. Listed by *Irving* [1964]. Ages were determined by *Allsopp* [1965] using Rb-Sr and K-Ar methods on minerals, Rb-Sr method on total rock samples, and a decay constant of 1.39×10^{-11} yr^{-1}. $N = 9$ sites.

1.3. *Great Dyke, Rhodesia* [*Nairn,* 1963]. Details are given by *Irving* [1964]. $N = 6$ sites.

1.4 *Gaberones granite, Botswana* [*Evans,* 1967]. A result from 7 sites (43 specimens from 19 cores) after magnetic cleaning of all specimens in fields of 50 to 1200 oe. One site was reversed with respect to the rest. Results were analyzed using the two-tier method of *Watson and Irving* [1957], k in Table 1 being the between-site precision so deduced. The baked-contact studies referred to in entry 1.1 provide strong evidence for stability. Rb-Sr age was determined by *McElhinny* [1966*a*] using total rock samples and a decay constant of 1.39×10^{-11} yr^{-1}.

1.5. *Gaberones granite, Botswana* [*Evans and McElhinny,* 1966]. A preliminary result from 2 sites (11 specimens from 4 cores) after magnetic cleaning in a field of 100 oe. These results were incorporated in a detailed study by *Evans* [1967], recorded in entry 1.4.

1.6 *Ventersdorp lavas, Cape Province* [*Jones et al.,* 1967]. Results from $N = 2$ sites (26 specimens from 13 hand samples) underground in the De

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Beers Mine at Kimberley, from 3 or possibly more separate flows. Magnetic cleaning in fields of 200 to 300 oe and thermal cleaning at 400°C gave results which were not significantly different. An age of about 2100 m.y. is suggested by *Nicolaysen* [1962].

1.7. *Bushveld gabbro, Transvaal* [*Gough and van Niekerk,* 1959]. Listed by *Irving* [1964]. $N = 5$ sites. The increase in precision on correcting for the pseudostratification observed in the gabbro is significant at the 95% confidence level [*McElhinny,* 1964]. Ages were determined by *Nicolaysen et al.* [1958] and *Schreiner* [1958*b*] and include Rb-Sr (decay constant 1.39×10^{-11} yr^{-1}) and U-Pb ages from total rock and separated mineral samples from the granite and U-Pb ages from a sample from the gabbro.

1.8. *Waterberg sandstones, Transvaal* [*Jones and McElhinny,* 1967]. Twelve sites (51 cores and 4 hand samples, 61 samples, 88 specimens) were investigated with large areal and stratigraphic coverage. All samples were subjected to magnetic cleaning in fields of 100 to 1800 oe. Samples from 6 sites were also thermally cleaned at 500 or 600°C, and generally no significant difference between the two methods was observed. Correction was made for the attitude of the beds. The results indicate that considerable polar wandering took place during the period of deposition of the sandstones. The polar-wander path so deduced is consistent with other Precambrian results for Africa.

The Waterberg system is younger than the Bushveld igneous complex (1950 ± 50 m.y., entry 1.7) and older than 1420 ± 70 m.y. There is a possibility that it may be older than 1790 ± 70 m.y. [*Oosthuyzen,* 1964].

1.9. *Mashonaland dolerite, Rhodesia* [*McElhinny and Opdyke,* 1964]. A recalculated result obtained from $N = 14$ sites (121 cores, 121 specimens, of which 5 were excluded from the analysis) after magnetic cleaning of all samples in fields of 100 to 800 oe. Four sites were reversed with respect to the rest. The authors report K-Ar age determinations on 3 samples which suggest an age of intrusion in excess of 1600 m.y.

1.10. *Premier Mine kimberlite, Transvaal* [*Jones,* 1968]. Seventeen hand samples were collected deep underground in the kimberlite pipe. One sample

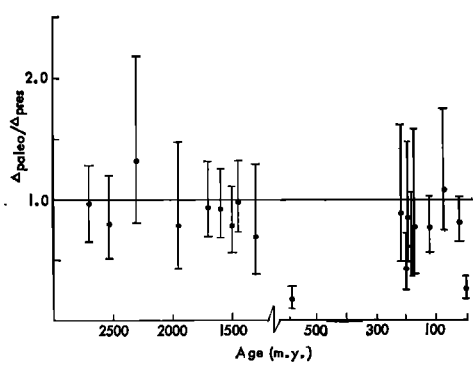


Fig. 10. Angular dispersions normalized to that of the present field and plotted as a function of time. Ninety-five per cent confidence limits are shown.

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was too weakly magnetized to be measured, and the result is based on $N = 16$ samples (32 specimens) after magnetic cleaning in a field of 300 oe. Five specimens were thermally cleaned at various temperatures up to 600°C, and no significant difference was found between the thermal and magnetic cleaning. The kimberlite is younger than the base of the Waterberg system and older than the Waterberg diabase sill which intrudes it. The sill has been dated [*Allsopp et al.*, 1967] at 1115 ± 15 m.y. by the Rb-Sr method on separated biotite, but incomplete studies on total rock and other separated minerals suggest a higher age. Lead model ages on galenas from the kimberlite suggest an age of 1750 ± 100 m.y. A preliminary result on the paleomagnetism of two samples from the pipe is given by *Jones and McElhinny* [1966].

1.11. *Waterberg diabases, Transvaal and Botswana* [*Jones and McElhinny*, 1966]. Result from $N = 13$ sites (90 samples from 62 cores and 5 hand samples) after magnetic cleaning of all samples in fields of 100 to 600 oe. One site was reversed with respect to the rest. The authors reported inconsistent K-Ar ages on total rock and separated minerals ranging from approximately 600 to 1300 m.y.

1.12. *Umkondo dolerites, Rhodesia* [*McElhinny and Opdyke*, 1964]. A recalculated result from $N = 9$ sites (77 cores, giving 77 samples, of which one was excluded from the final analysis) after magnetic cleaning of all samples in fields of 100 to 1000 oe. In the original paper 10 sites were reported, but one of these was subsequently found to be a lava and is included in entry 1.13. *Jones and McElhinny* [1966] report inconsistent K-Ar ages from 3 total rock samples and 1 separated mineral fraction ranging from approximately 650 to 1150 m.y.

1.13. *Umkondo lavas, Rhodesia* [*McElhinny*, 1966*b*]. Results from $N = 9$ sites (21 cores) after magnetic cleaning at all sites in fields of 200 to 800 oe. Calculations by the present authors based on the original data give a slightly different result (1.13) from that quoted in the original.

1.14 *Umkondo lavas and dolerites combined* [*McElhinny*, 1966*b*]. A combination of entries 1.12 and 1.13 recalculated from the original data.

1.15. *Pilansberg dikes, Transvaal* [*Gough*, 1956]. Listed by *Irving* [1964]. $N = 5$ sites (dikes). Ages are based on Rb-Sr measurements by *Schreiner* [1958*a*] and *van Niekerk* [1962], who used a decay constant of 1.39×10^{-11} yr^{-1}.

1.16 *Fort Gouraud iron ore, Mauretania* [*Gross and Strangway*, 1961]. Fifty-six samples collected from two limbs of the F'derick ore body in the Fort Gouraud deposits show consistent direction after restoring the limbs to the horizontal and rotating one of them into alignment with the other. No paleomagnetic direction can be inferred from the result because it is not known which of the two limbs is in its original attitude.

1.17. *Van Dyk Mine dolerite dike* [*Jones and McElhinny*, 1966]. Results from $N = 8$ hand samples (16 specimens) after magnetic cleaning in a field of

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250 oe. The samples were taken at a depth of 1400 meters below the surface and were uniformly spread over the 30-meter width of the dike. Its direction of magnetization does not correlate with any of the other Precambrian results and the result is bracketed. *McDougall* [1963] reports a minimum age of 1120 m.y. for this dike from K-Ar measurements on separated feldspar.

APPENDIX 2. NOTES ON TABLE 2

2.1. *Lavas, east-central Morocco* [*Helsley,* 1965]. An unspecified number of samples treated in alternating magnetic fields up to 5000 oe. Normal and reversed directions (174° apart) were detected. In the absence of sufficient geological and sampling information, the result is regarded as tentative and is bracketed.

2.2 *Ntonya ring structure, Malawi* [*Briden,* 1968]. Mean of 7 sites after magnetic cleaning in fields up to 800 oe. Four other sites, being scattered, were rejected. Thermal demagnetization confirmed the presence of the stable component. The age is based upon K-Ar age determination by N. J. Snelling quoted by *Briden* [1968].

2.3. *Hook intrusives, Zambia* [*Brock,* 1967]. Mean of stable directions obtained at 3 sites after magnetic cleaning in fields up to 200 oe. Other sites gave scattered results. All sites are in gabbro and diorite satellite bodies which are post-tectonic and intrude folded Kundelungu sediments. A red granite, also believed to be post-tectonic [*De Swart and Drysdall,* 1964] has been dated [*Snelling et al.,* 1964] at 500 ± 17 m.y. using the Rb-Sr method on total rock and biotite samples and a decay constant of 1.475×10^{-11} yr^{-1}, K-Ar ages on biotite give 465 ± 25 m.y. The Rb-Sr age is preferred by Snelling et al.

2.4. *Dedza Mountain syenite, Malawi* (J. C. Briden, unpublished result). Two cores were drilled at one site on top of the mountain. After magnetic cleaning to remove large soft components, one core was not measurable. The other gave the direction quoted here, which is very similar to that of the Ntonya ring structure (entry 2.3) and supports the geological idea that the two bodies are of similar age.

2.5. *Table Mountain Series, Cape Town* [*Graham and Hales,* 1961]. Result based on 8 cores (107 specimens) from a 2-foot red siltstone band in a road cutting. Results from natural outcrops were scattered and were not used in the analysis. *Graham et al.* [1964] reported that the stability of these samples has been tested by alternating magnetic fields, but no details were given. Although this is only a small collection, which is unlikely to cover secular variation, the direction probably approximates to that in Silurian times in southern Africa.

2.6. *Table Mountain Series, Natal* (J. C. Briden and A. Brock, unpublished result). Eight sites were sampled in thin red siltstone bands, in sequences correlated with the T.M.S. of the Cape Province, in the Durban-Kranskop district. The 46 specimens measured gave directions slightly shallower than the present field inclination ($D = 340½$, $I = -64½$) before being corrected for

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the tilting of the beds. Thermal demagnetization did not reveal any underlying stable component, and the NRM is likely to be of recent origin. The result is therefore bracketed.

2.7, 2.8. *Dwyka varved clay, Rhodesia* [*Nairn,* 1960]. A total of 4 samples yielded normal direction ($N = 10$ specimens, entry 2.7) and reversed directions ($N = 9$ specimens, entry 2.8). The result is bracketed because of the small number of samples collected and the absence of stability tests apart from a reversal.

2.9. *Dwyka varves, central Africa* [*McElhinny and Opdyke,* 1968]. Fifty-two samples were collected at 5 sites in glacial varves in the Galula coalfield, Tanzania (2 sites), southern Zambia (2 sites), and northwestern Rhodesia (1 site, the same locality as entries 2.7 and 2.8). After thermal cleaning in temperatures of 300 to 450°C, only 23 samples still could be measured. These gave well-grouped directions at each site (4 reversed, 1 normal). If the beds are restored to the horizontal, the remanence converges to a steeply inclined direction. The Dwyka series is normally regarded as being Late Carboniferous in age, but little or no fossil evidence from these sites substantiates this supposition. The authors discuss a possible Early Carboniferous age for these varves.

2.10, 2.11, 2.12. *Permian redbeds, Tanganyika* [*Opdyke,* 1964*b*]. The K3 beds in the Songwe-Kiwira and Ketewaka-Mchuchuma coalfields are of Early Permian age (entry 2.10). The K3 beds of the Galula coalfield are not so precisely dated and give a significantly different pole (entry 2.11). This pole is more compatible with a Late Carboniferous age if the polar-wander curve presented in this paper is correct. These poles have been calculated from the published data and are also given by *McElhinny and Opdyke* [1968]. Entry 2.10 ($N = 4$ sites, 27 samples), entry 2.11 ($N = 5$ sites, 34 samples), and entry 2.12 ($N = 9$ sites) give the combined result recalculated and slightly different from the original published data.

2.13. *Ecca redbeds, Songwe-Kiwira* [*Nairn,* 1964]. A collection from the same site as some of the samples in entry 2.10. Results from 11 disks were selected either because they had NRMs farthest from the present field direction or because they gave this direction after magnetic cleaning in fields of 750 to 1000 oe. The result is bracketed because the collection is small, but it is a useful confirmation of the result in entry 2.10.

2.14. *Taru grit* [*Creer et al.,* 1958; *Nairn,* 1960]. $N = 32$ specimens from 8 samples. No magnetic cleaning was attempted and the result is therefore bracketed.

2.15. *Maji Ya Chumvi formation, Kenya* [*Creer et al.,* 1958; *Nairn,* 1960]. $N = 21$ specimens from 5 samples. It is stated of this and the Taru grit (entry 2.14) [*Nairn,* 1960] that 'one fifth of the samples appeared to be stable' on the basis of an unspecified fold test. *Cox and Doell* [1960] point out that the statistical values are internally inconsistent and the means of deriving them are obscure. The result is bracketed.

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2.16. *Microdiorite dike, Monkey Bay, Malawi* (J. C. Briden, unpublished result). Four cores from a single dike were demagnetized in fields up to 450 oe. Magnetic cleaning was achieved at 300 oe. The result merely confirms the suspected Karroo age of the dike. The pole position and reversed polarity are compatible with a Late Carboniferous to Early Permian age, but this conclusion is tentative. The sample is too small and the result is bracketed.

2.17. *Série rouge inférieure, Madagascar* [*Nairn,* 1964]. Results of total NRM measurements on 15 specimens from 6 samples at 2 sites. The confidence limits of the mean direction and pole are incompatible with the stated precision. For $N = 15$, $k = 13$, α_{95} is approximately 10°. The result is bracketed.

2.18. *Shawa ijolite, Rhodesia* [*Gough and Brock,* 1964]. Result from $N = 5$ sites (13 cores, 48 specimens) after magnetic cleaning in fields of 150 to 400 oe. Age based upon Rb-Sr method and a decay constant of 1.386×10^{-11} yr^{-1} [*Nicolaysen et al.*, 1962].

2.19. *Upper Beaufort sediments, South Africa* [*Graham,* 1961]. Result from 19 cores from 4 sites spread over more than 150 km in the vicinity of Queenstown. According to *Graham et al.* [1964], 'The direction of remanent magnetization of these cores remains unchanged after magnetic cleaning in fields up to 550 oersteds.' In the absence of any properly published information, this result has been bracketed. No statistics are published—only the pole.

2.20. *Red shale, Middle Beaufort, Natal* [*Nairn,* 1964]. Inital measurements on 7 specimens from 2 samples. The result is entered in brackets.

2.21. *Stormberg redbeds, Orange Free State* [*Nairn,* 1964]. Initial measurements on 14 specimens from 5 samples at 1 site. Result entered in brackets.

2.22. *Sandstones, Rhodesia and Zambia* [*Nairn,* 1960]. $N = 6$ sites (99 specimens from 21 samples). Two sites are in escarpment grit, 2 in red sandstone formation, and 2 in Forest sandstone. No cleaning was done, and the direction is not significantly different from the present field direction at the sampling sites; therefore the result is bracketed. The pole is quoted from the original; recalculation using the published data indicates that it might more correctly be 82°S, 124°E, but the discrepancy is small.

2.23. *Red sandstone, Songwe rift valley* [*Nairn,* 1964]. $N = 11$ specimens from 5 samples at 1 site in K5 sandstones of Late (?) Triassic age. Magnetic cleaning in fields up to 750 oe did not alter directions significantly. This preliminary result does not agree with other Triassic results, and, in view of the small number of samples collected, the result is entered in brackets.

2.24. *Red sandstone formation, Zambia* [*Opdyke,* 1964*a*]. $N = 6$ sites out of 15 which were sampled. The others were too weakly magnetized to be measured. Thirty-two cores were used in the analysis. Magnetic cleaning did not alter the NRM direction or intensity except at one site, where magnetic cleaning was achieved in a field of 800 oe. Blocking temperatures were distributed between room temperature and the Curie point of hematite (670°C).

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2.25. *Cave sandstone, Botswana* [*Nairn,* 1960]. $N = 52$ specimens from 10 samples. No cleaning was attempted. The small errors quoted are due to measuring a large number of specimens from the samples. The result is bracketed.

2.26. *Nuanetsi lavas, Rhodesia* [*Brock,* 1968]. Result based on 6 samples after magnetic cleaning. The precision from this small collection is low, and its main significance lies in the reversed polarity. It is entered in brackets. Age is based on Rb-Sr measurements (W. Manton, personal communication) on rhyolites which overlie the lavas, using a decay constant of 1.39×10^{-11} yr^{-1}.

2.27 *Stormberg lavas, Lesotho,* collection of J. S. V. van Zijl [*Hales,* 1960]. This preliminary result is based on NRM measurements of 119 cores at 14 sites and is superseded by results of 2.28–2.32.

2.28–2.32. *Stormberg lavas, Lesotho* [*van Zijl et al.,* 1962*a,b*]. Summarized by *Irving* [1964]. Unit weight is given to samples in entries 2.28–2.31 ($N = 33$, 13, 21, and 7, respectively). Entry 2.32 reproduces *Irving*'s [1964] calculation, in which unit weight is given to each of the entries 2.28–2.31 and reversal is considered. Because normal and reversed polarities are equally represented, columns 5 and 6 could equally well read '$D = 158$, $I = +56$.' In column 16, the pole is designated N/S. The lavas are considered to be essentially the same age as the Karroo dolerites (2.36–2.44), which have been dated radiometrically.

2.33. *Batoka basalts, Rhodesia* [*Nairn,* 1960]. Three out of 4 flows at Victoria Falls and 1 sample from Bulawayo district are used in the analysis ($N = 44$ specimens from 12 samples). The remainder of the collection was discarded as randomly magnetized. No cleaning was attempted. The above reference gives a slightly amended mean direction from *Nairn* [1956, 1957].

2.34. *Karroo lavas, Rhodesia and Mozambique* [*Opdyke,* 1967]. Results from 5 lava flows are reported. Three flows are from Bulawayo, 1 from Gokwe in Rhodesia, and 1 from Sungo in Mozambique. Flow BK1 from Bulawayo gives an 'intermediate' direction which can be correlated with the intermediate zone during the reversal seen in the Stormberg lavas (entries 2.28–2.31). The results from the remaining 4 flows are incorporated in entry 2.35. All samples were magnetically cleaned in fields up to 200 oe.

2.35. *Karroo lavas, central Africa* (a combined result). A combination of results from the 5 flows of entry 2.33 and 4 flows of entry 2.34, calculated by giving unit weight to each flow.

2.36. *Karroo dolerites, surface collections* [*Graham and Hales,* 1957]. Result based on $N = 33$ samples from 20 sites, 8 samples being reversely magnetized. No cleaning was attempted, but sites from which samples acquired VRM in the laboratory were excluded. Because of this and the large dispersion, the result is of minor importance compared with those of entries 2.37–2.44 and 2.48. K-Ar age determinations by *McDougall* [1963] were in the range 190–154 m.y.

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2.37. *Karroo dolerite sill, Umgeni valley* [*Nairn,* 1964]. Eleven specimens from 3 samples were magnetically cleaned in a field of 250 oe. The result is given unit weight in entry 2.48.

2.38. *Limburgite plug, Rhodesia* [*Nairn,* 1964]. Seven specimens from 2 samples at Guinea Fowl siding near Gwelo. No cleaning was attempted, but the reversed direction indicates some stability and the result is given unit weight in entry 2.48.

2.39. *Karroo dolerites, Rhodesia* [*McElhinny and Jones,* 1965]. Four sites with normal polarity in the eastern districts of Rhodesia were subjected to magnetic cleaning. Results are incorporated in entry 2.48, unit weight being given to each site.

2.40–2.44. *Karroo dolerites, mines and tunnels* [*Graham and Hales,* 1957]. Results of total NRM measurements, unit weight being given to samples (entries 2.40–2.43) recalculated by *Cox and Doell* [1960] from published data. Entry 2.44 is that given by Cox and Doell, unit weight being given to each of the entries 2.40–2.43; both polarities are equally represented, hence the designation N/S in column 16. Avoidance of surface weathering and the presence of a reversal provide some evidence for stability. Entries 2.40–2.43 are given unit weight in entry 2.48.

2.45–2.46. *Baked Karroo sediments, Estcourt* [*Graham and Hales,* 1957]. These samples were baked by the normal and reversed dolerites at Estcourt. 2.45: $N = 7$ samples (24 specimens). 2.46: $N = 8$ samples (25 specimens). No cleaning was attempted, but the agreement of polarity and direction with adjacent dolerites constitutes a contact test of stability of the NRM in those dolerites. Each entry is given unit weight in 2.47.

2.47. *Karroo dolerites and baked sediments from mines and tunnels combined* [*Graham and Hales,* 1957]. A recalculation by *Irving* [1964] gave unit weight to each of the entries 2.40–2.43, 2.45, 2.46 and took account of reversals. Both polarities are equally represented; the polarity of the quoted direction is therefore arbitrary (columns 5 and 6) and the pole is designated N/S in column 16. An error in the longitude of the pole has been corrected here.

2.48. *Karroo dolerites combined* [*McElhinny and Jones,* 1965]. A calculation in which unit weight is given to each of the entries 2.37, 2.38, 2.40–2.43, and each of the 4 sites in 2.39. $N = 10$, of which 3 are reversely magnetized.

2.49. *Marangudzi ring complex, Rhodesia* [*Gough et al.,* 1964*a*; *Brock,* 1968]. Results from 8 sites after magnetic cleaning. Seven sites were reversed and 1 is normal. The original reference reported results from 5 sites, but this entry now includes 8 sites. All samples were magnetically cleaned in fields up to 800 oe. K-Ar age determinations by J. A. Miller quoted by the original authors range from 182 ± 10 to 196 ± 10 m.y. Six samples were dated (4 biotites, 1 hornblende contaminated with felspar, and 1 granite total rock).

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2.50. *Mateke Hills ring complexes* [*Gough et al.,* 1964*a*]. Results from 6 sites after magnetic cleaning in fields of 100 to 500 oe. A seventh site with 'normal' polarity (8 cores, $D = 299$, $I = 30$, $k = 31$, $\alpha_{95} = 10°$) has been excluded because its mean direction lies about 145° from the mean reversed direction. Age is based on Rb-Sr measurements by W. Manton (personal communication), who used a decay constant of 1.39×10^{-11} yr^{-1}.

2.51. *Red siltstone, Mozambique* [*Gough and Opdyke,* 1963]. This is correlated with the Lupata series (Lower Cretaceous; see entry 2.52). The direction is quoted after correction for 5°W dip. $N = 5$ hand samples were magnetically cleaned in a field of 100 oe. Pole was calculated by *Irving* [1964] from published data.

2.52. *Lupata series volcanics, Mozambique* [*Gough and Opdyke,* 1963]. Results from $N = 7$ sites (6 or 7 lava flows) after correction for geological dip and magnetic cleaning in fields of 150 to 500 oe. α_{95} calculated by *Irving* [1964] from published data. K-Ar ages determined by McDougall (quoted by *Gough et al.* [1964*a*]) on anorthoclase are 106.7 and 110.5 m.y.

2.53. *Mlanje Massif* [*Briden,* 1967*a*]. Results from $N = 8$ samples over 100 meters across outer syenite ring, after magnetic cleaning in fields up to 450 oe. K-Ar analysis by N. J. Snelling (personal communication) gave 116 ± 6 and 128 ± 6 m.y.

2.54. *Lupata series volcanics and Mlanje Massif combined* [*Briden,* 1967*a*]. Combined result giving unit weight to each sample in entry 2.53 and to each site in entry 2.52.

2.55. *Red sandstone, Songwe River* [*Nairn,* 1964]. A preliminary result, no attempt at cleaning having been made, from $N = 11$ specimens from 3 samples. It is entered in brackets.

2.56. *Lavas and dikes, Madagascar* [*Roche et al.,* 1958; *Roche and Cattala,* 1959]. The lavas are stated to be Cenomanian to Santonian. Ten sites are spread over 1000 km, and site mean directions are given in the original. Thermal demagnetization indicated stability and a single Curie point.

2.57. *Lavas, Madagascar* [*Nairn,* 1964]. Twelve specimens from 3 sites in 2 lavas. The sites are probably the same as some in the preceding entry. The optimum field for magnetic cleaning was 250 oe. Scatter increased considerably at 300 oe. The result is in rough agreement with entry 2.56. (The pole is quoted from the original; recalculation from the published mean direction yields a pole at 49°S, 8°W.)

2.58. *Ethiopian traps* [*Grasty,* 1964]. The original result was quoted for $N = 21$ samples from 11 sites, giving $D = 3$, $I = +14$, $\alpha_{95} = 7°$ from which *Raja et al.* [1966] calculate a pole at 86°S, 94°W, with $dp = 3½$, $dm = 7½$, but there seems to be an error in this calculation. The result given here is recalculated by the authors from Grasty's data, using $N = 11$, which is the number of sites and is probably the number of flows. Stability is confirmed by thermal

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demagnetization. K-Ar age determinations on 5 of the paleomagnetic samples gave ages ranging from 30 to 69 m.y. [*Grasty et al.*, 1963]. The estimates of younger ages may be due to alteration (chloritization?). The greatest age is from the sample with highest K content and lowest atmospheric Ar contamination, and it is likely to be the best estimate. The age is therefore given here as Late Cretaceous to Eocene.

2.59. *Turkana lavas* [*Raja et al.*, 1966]. This supersedes a preliminary result given by *Mussett et al.* [1964]. The analysis is based on samples ($N = 57$) because the within-sample dispersion was as large as between-sample and between-site dispersions (personal communication from the original authors). Specimens were magnetically cleaned in fields up to 1300 oe and some specimens were also thermally demagnetized. About two-thirds of the collection was normally magnetized. K-Ar analyses on 4 samples [*Reilly et al.*, 1966] ranged from 12.5 to 32.2 m.y., and the greater ages are likely to be the better estimates.

2.60. *Cuirasse ferrugineuse, Madagascar* [*Nairn*, 1964]. A preliminary result, no attempt at cleaning having been made, from $N = 12$ specimens (1 sample at each of 4 sites) is entered in brackets.

2.61. *Lavas, Nairobi district* [*Nairn*, 1964]. $N = 6$ sites in different flows (5 normal, 1 reversed). 'In most cases two sites per flow were visited and two or three samples per site were collected.' Magnetic cleaning was done in fields of 50 to 150 oe. Nairn quotes all these but one to be Miocene, but *Raja et al.* [1966] suggest Plio-Pleistocene age. *dp, dm* were calculated by Raja et al. from published data.

2.62. *Rungwe volcanics, Tanzania* [*Nairn*, 1964]. Result based on $N = 9$ samples from 3 flows (2 normal, 1 reversed) calculated by *Raja et al.* [1966] using published data; an error in their calculation of the pole has been corrected here. Magnetic cleaning was done in a field of 250 oe.

2.63. *Rift Valley lavas, Kenya* [*Nairn*, 1964]. Result based on $N = 11$ samples (27 specimens) from 4 lava flows, all reversed. Magnetic cleaning was done in a field of 250 oe. Calculation by *Raja et al.* [1966] using published data; an error in their calculation of the pole has been corrected here.

2.64. *Rungwe and Kenya Rift Valley lavas combined* [*Nairn*, 1964]. Combination of entries 2.62 and 2.63 in which unit weight is given to each *flow* ($N = 7$; 2 normal and 5 reversed) resulting in a pole slightly to the west of those given in 2.62 and 2.63.

2.65. *Lavas, south rift, Kenya* [*Mussett et al.*, 1964]. Initial measurements on $N = 83$ samples (35 sites in about 22 flows) are reported; an error in the original calculation of the pole has been corrected here. Some of the data were rejected on the basis of within-site scatter. The majority of the collection had normal polarity. In the absence of rigorous stability tests the result is bracketed.

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2.66. *Bed I, Olduvai Gorge, Tanzania* [*Grommé and Hay,* 1963, 1967]. Calculation by the present authors from published data, unit weight given to each lava flow ($N = 2$). *Grommé and Hay* [1963] report $D = 5°$, $I = +½°$, $k = 220$; they gave unit weight to 12 specimens from 4 samples in the upper (pahoehoe) flow after magnetic cleaning (100 oe) and thermal cleaning (250°C). In the lower (aa) flow $N = 10$ cores gave $D = 359°$, $I = -7½°$, $k = 267$; scatter was greatly increased by demagnetization in 100-oe peak field [*Grommé and Hay,* 1967]. The age of Bed I is shown to be 1.9 m.y. by *Evernden and Curtis* [1965].

2.67. *Aden volcanics* [*Irving and Tarling,* 1961]. Results of magnetic cleaning in fields up to 900 oe are analyzed by the *Watson and Irving* [1957] two-tier method on 54 samples from 12 sites (3 normal, 9 reversed). The between-site precision is given here. The *Fisher* [1953] precision on site mean directions is 96 [*Briden and Ward,* 1966].

2.68. *Ahaggar basalts* [*Roche and Leprêtre,* 1955]. Calculation by *Irving* [1964] for $N = 6$ flows (22 samples from 2 volcanic series); directions are quoted in the original.

2.69. *Archaeological baked clays* [*Thellier and Thellier,* 1951]. Initial measurements on 2 Punic kilns (~140 B.C.) and 1 Roman hearth (~300 A.D.); recalculated by *Irving* [1964], who gave unit weight to each age ($N = 2$).

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